

IdeenExpo Hannover Demo Guide

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This repository contains several small 42-style projects prepared for live demos at IdeenExpo in Hannover. The goal is to make the projects easy to compile, run, and tweak while visitors are watching.

Most C graphics projects use the shared libraries in `common/`:

- `common/minilibx-linux/` for classic MiniLibX projects.
- `common/mlx42/` for MLX42 projects.
- `common/libft*` for shared libft variants used by multiple projects.

Keep those shared folders in place. They avoid having a separate MiniLibX or libft copy inside every project.

General Workflow

For most C projects:

```
cd path/to/project
make
./program_name path/to/map
```

Common Makefile targets:

- `make`: build the normal executable.
- `make bonus`: build the bonus version when the project has one.
- `make clean`: remove object files.
- `make fclean`: remove object files and executables.
- `make re`: rebuild from scratch.

If you only edit a map or config file, usually you do not need to recompile. If you edit `.c` or `.h` files, run `make re`.

Quick Run Commands

Run commands from the project directory shown in the left column.

Project	Build	Run example
<code>a-maze-ing/ilobov_myaroshu</code>	<code>make install</code>	<code>make run</code>
<code>a-maze-ing/kaggour_jbuchins</code>	<code>make install</code>	<code>make run</code>
<code>so_long/nradin</code>	<code>make</code>	<code>./so_long maps/map_1.ber</code>
<code>so_long/nradin bonus</code>	<code>make bonus</code>	<code>./so_long maps/map_1_bonus.ber</code>
<code>so_long/jastomme</code>	<code>make</code>	<code>./so_long map/minimal.ber</code>
<code>so_long/pmichale</code>	<code>make</code>	<code>./so_long mini.ber</code>
<code>cub3d/triedel_macruz-sa</code>	<code>make</code> or <code>make bonus</code>	<code>./cub3D lvl/level0.cub</code> or <code>./cub3D_bonus lvl/level0.cub</code>

Project	Build	Run example
cub3d/alappas	make or make bonus	./cub3D maps/map1.cub or ./cub3D_bonus maps/map_bonus.cub
cub3d/mperetia_dyarkovs	make or make bonus	./cub3D maps/map.cub
cub3d/robello_gkamanur	make or make bonus	make info shows available maps and targets
fdf/triedel	make	make run or ./fdf m maps/42.fdf
fdf/dyarkovs	make	make run
fractol/cjaks	make	./fractol 1 1
fractol/loandrad	make	./fractol mandelbrot or ./fractol julia
push_swap/*	make	Use the Godot visualiser; see the Push Swap section below

If a map path is missing in one project, list the available maps:

```
find . -name "*.ber" -o -name "*.cub" -o -name "*.fdf"
```

Simple Tweaks For Visitors

1. Change A Maze

For a-maze-ing projects, edit config.txt.

Example:

```
WIDTH=20
HEIGHT=15
ENTRY=0,0
EXIT=19,14
OUTPUT_FILE=maze.txt
PERFECT=True
SEED=4242424242
```

Good demo tweaks:

- Change WIDTH and HEIGHT for a bigger or smaller maze.
- Change ENTRY and EXIT to move the start and goal.
- Change SEED to get a different maze while keeping it reproducible.

Then run:

```
make run
```

2. Change A so_long Map

Maps use simple characters:

- 1: wall
- 0: floor
- P: player start
- E: exit
- C: collectible
- H or V: enemies in bonus maps, for projects that support them

Example from `so_long/nradin/maps/map_1_bonus.ber`:

```
111111
1EH001
101C01
100V01
1P0HC1
111111
```

Rules to keep:

- The map must be rectangular.
- The outside border should be walls.
- Keep exactly one P and one E.
- Keep at least one C.
- Use `make bonus` before running maps with H or V in `so_long/nradin`.

3. Change A cub3D Map

Cub3D maps are `.cub` files. The top part configures textures and colors; the bottom part is the maze.

Common entries:

```
N0 textures/color_stone.xpm
S0 textures/color_stone.xpm
EA textures/color_stone.xpm
WE textures/color_stone.xpm
```

```
F 0,0,0
C 200,200,200
```

```
111111
1000N1
111111
```

Safe demo tweaks:

- Change the maze layout by editing 1 walls and 0 floor tiles.
- Move the player by moving N, S, E, or W.
- Change floor and ceiling colors with `F R,G,B` and `C R,G,B`.

Do not replace or edit sprite/texture image files during the event. Many MLX projects expect exact image dimensions, and a mismatched `.xpm` or `.png` can break the demo. If a wall should look different, prepare the correctly sized asset before the event rather than doing it live.

4. Change An FdF Shape

FdF maps are grids of numbers. Bigger numbers make higher points.

Example:

```
0 0 0 0 0
0 1 2 1 0
0 2 4 2 0
0 1 2 1 0
0 0 0 0 0
```

Good demo tweaks:

- Raise one number to create a mountain.
- Use negative numbers to make a valley.
- Add colors in the standard FdF style when supported: 10,0xFF0000.

Run examples:

```
cd fdf/triedel
make
./fdf m maps/42.fdf
```

```
cd fdf/dyarkovs
make
make run
```

5. Change Fractol Feel

For fractol/cjakobs, the zoom speed is intentionally easy to tweak:

```
zoom_step = 0.08;
```

File:

```
fractol/cjakobs/src/navigation.c
```

Smaller values feel smoother but require more scroll input. Larger values zoom faster but can feel jumpy.

Run:

```
cd fractol/cjakobs
make
./fractol 1 1
```

The first number chooses the fractal type, and the second chooses a color palette.

6. Run Push Swap With The Visualiser

For push_swap, do not demo it by reading the terminal output directly. Use the Godot visualiser.

Build one of the push_swap projects first:

```
cd push_swap/radix_lflorin
make
```

Then start the visualiser:

```
cd ../Visualiser
./PushSwapVisualiser.x86_64
```

In the visualiser:

- Drag and drop the compiled ./push_swap executable into the visualiser.
- Press the automation button.
- Let the visualiser run the executable and animate the operations.
- Repeat it several times for each algorithm and let them run in the background

The same workflow works for the other push_swap/* implementations after they are compiled.

Good Expo Demo Ideas

- Ask a visitor to draw a tiny so_long level using 1, 0, P, E, C.
- Let someone change F and C in a .cub file and immediately rerun cub3D.
- Change a maze SEED and show how the same seed creates the same maze again.
- In FdF, change a single height value and show the terrain spike.
- In Fractol, change zoom_step, rebuild, and compare how zooming feels.
- In Push Swap, drag a compiled executable into the visualiser and run the automation.

Troubleshooting

If make cannot find MiniLibX, check that the shared library folder exists:

```
ls common/minilibx-linux
```

If MLX42 projects fail, check:

```
ls common/mlx42
```

If a graphical program opens no window, make sure you are running in a graphical Linux session, not a plain SSH terminal without display forwarding.

If a so_long bonus map says it contains invalid characters, rebuild the bonus version first:

```
make bonus  
./so_long maps/map_1_bonus.ber
```

If you get lost during a live demo, return to a known-good state:

```
make fclean  
make
```

Then run one of the examples from the table above.